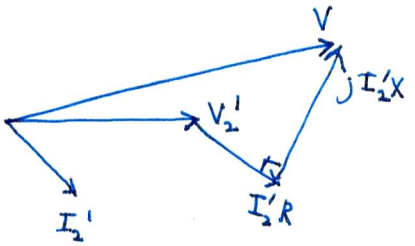
① 200V/400V, 50Hz, 4 kVA transformer.

Referring to LV side (1° side): $R = 0.15\Omega$ $X = 0.37\Omega$; $R_m = 600\Omega$; $X_m = 300\Omega$

Load data: 10A current at 0.8 pf lagging (at HV side)

I_2' (ref. to LV side) $= 10 \times \frac{400}{200} = 20A$

From phasor diagram.



$V \approx V_2' + I_2'(R\cos\phi + X\sin\phi)$

$\Rightarrow V_2' \approx V - I_2'(R\cos\phi + X\sin\phi)$

Here wrt LV side, $V = 200V$; $R = 0.15\Omega$; $X = 0.37\Omega$; $|I_2'| = 20A$

$\cos\phi = 0.8$

$\Rightarrow V_2' \approx 200 - 20(0.15 \times 0.8 + 0.37 \times 0.6)$

$= 193.16\,V$

High voltage side terminal voltage $= V_2 = V_2'\left(\frac{N_2}{N_1}\right) = 193.16 \times \frac{400}{200}$

$= 386.23\,V$

Low voltage side input current

$\bar{I}_1 = \bar{I}_{sh} + \bar{I}_2'$

$\bar{I}_2' = 20\angle -36.86^\circ\ A$

$\bar{I}_{sh} \cong \frac{200\angle 0}{600\Omega \parallel j300\Omega} = \frac{200\angle 0}{\frac{600 \times j300}{600 + j300}} = 0.7454\angle -63.4349^\circ\ A$

$\bar{I}_1 = 20.6692\angle -37.7844^\circ\ A$.

Efficiency.

output power $= V_2' I_2' \cos\phi = 193.16 \times 20 \times 0.8 = 3090.56\,W$

Copper loss $= (I_2')^2 R = 20^2 \times 0.15 = 60\,W$.

Iron loss $= \frac{V^2}{R_m} = \frac{200^2}{600} = 66.66\,W$.

$\eta = \left(\frac{3090.56}{3090.56 + 60 + 66.66}\right) = 96.06\%$

2) 25 kVA, 2000 V/200 V

Iron loss = 350 W

Copper loss = 400 W (at full load)

a) (i) Full load at Unity Power Factor.

Output power = $25 \times 10^3 \times 1$ = 25000 W

Iron loss = 350 W

Copper loss @ full load condition = 400 W

Input power = 25000 + 350 + 400 = 25750 W

Efficiency = $\frac{\text{Output power}}{\text{Input power}}$ = 97.087 %

(ii) Half full load at UPF (supplies half the rated VA @ UPF)

Output power = $0.5 \times 25 \times 10^3 \times 1$ = 12500 W

Iron loss = 350 W

Copper loss at half the full load condition = ?

suppose I_{fl} is the full load current

$x\ I_{fl}$ is the current at x times full load

Copper loss at Full load = $P_{cufl} = I_{fl}^2 R$

Copper loss at x times rated load = $P_{cu\,xfl} = x^2 I_{fl}^2 R$

$\therefore \frac{P_{cuxfl}}{P_{cufl}} = x^2 \Rightarrow P_{cu_xfl} = x^2 . P_{cufl}$

$\therefore$ Copper loss at $(\frac{1}{2})$ the rated load = $(\frac{1}{2})^2$ * Copper loss at rated load

= $0.5^2 * 400$ = 100 W

Input power = 12950 W

Efficiency = 96.525 %

b) suppose max. η occurs at x times the rated load

$\Rightarrow$ ~~$x I_{fl}$~~ = $\sqrt{\frac{\text{Iron loss}}{R}}$ = $\sqrt{350}$

$\Rightarrow$ For maximum efficiency, Iron loss = Copper loss

$\Rightarrow x^2 I_{fl}^2 R$ = Iron loss

$\Rightarrow x^2 = \frac{\text{Iron loss}}{I_{fl}^2 R} = \frac{\text{Iron loss}}{\text{Copper loss @ FL}} \Rightarrow x^2 = \frac{350}{400}$

$x = 0.93541$

$\Rightarrow$ max. η occurs at 93.54 % of rated load.

Iron loss = Copper loss = 350W at that condition.

③ 10kVA, 50Hz, 2200V/220V

$R_1 = 4\Omega$ $R_2 = 0.04\Omega$ $X_1 = 5\Omega$ $X_2 = 0.005\Omega$

a) Leakage impedance referred to HV side (1° side)

$R = R_1 + R_2' = 4 + 0.04 \times \frac{2200^2}{220^2} = 8\Omega$

$X = X_1 + X_2' = 5 + 0.005 \times \frac{2200^2}{220^2} = 5.5\Omega$

$\Rightarrow Z = (8 + j5.5)\Omega$

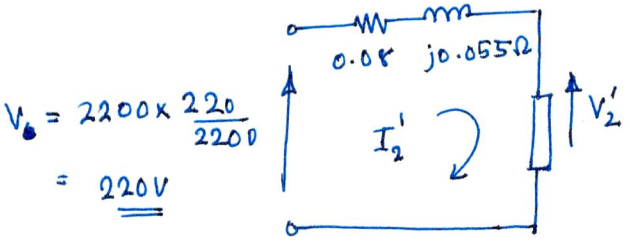
Leakage impedance referred to LV side (2° side)

$R = R_1' + R_2 = 4 \times \frac{220^2}{2200^2} + 0.04 = 0.08\Omega$

$X = X_1' + X_2 = 5 \times \frac{220^2}{2200^2} + 0.005 = 0.055\Omega$

$\Rightarrow Z = (0.08 + j0.055)\Omega$

b) Equivalent circuit referring to LV side.



Flow of current in LV side

$\Rightarrow$ Rated current at 0.8 pf lag.

Rated current $= \frac{10 \times 10^3}{220}$

$= 45.4545\ A$.

$\bar{I}_2' = 45.45 \angle -36.86°$

Voltage across load terminals, $|\bar{V}_2'| = |V| - |I_2'|(R\cos\phi + X\sin\phi)$

$= 220 - 45.45(0.08 \times 0.8 + 0.055 \times 0.6)$

$= 215.5913\ V$ $\Rightarrow$ % regulation $= \frac{220 - 215.5913}{220} = 2\%$

c) Leading pf 0.8

$|V_2'| = |V| - |I_2'|(R\cos\phi - X\sin\phi)$

$= 218.59\ V$

$\Rightarrow$ % regulation $= \frac{220 - 218.59}{220} = 0.64\%$

d) Output power $= |V_2'||I_2'|\cos\phi$

For lagging load, Output power $= 215.5913 \times 45.45 \times 0.8$

$= 7838.8997\ W$

Iron loss = 80W

Copper loss $= (I_2')^2 R = 45.45^2 \times 0.08 = 165.2562\ W$.

Input power $= 8084.1559\ W$.

Efficiency = 96.96%

Now for leading load,

Output power = $218.59 \times 45.45 \times 0.8 = 7947.9324$ W

Iron loss = 80 W

Copper loss = $45.45^2 \times 0.08 = 165.2562$ W.

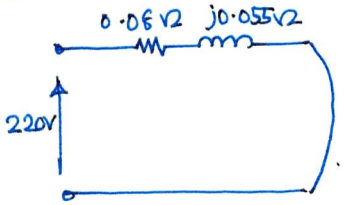
Input power = 8193.1886 W

$\Rightarrow$ Efficiency, η = 97%

(e) short circuit current

$$I_2' = \frac{220}{0.08 + j0.055}$$

$$= 2266.1153 \text{ A}$$



short circuit current in HV side = $2266.1153 \times \frac{220}{2200} = 226.61$ A.

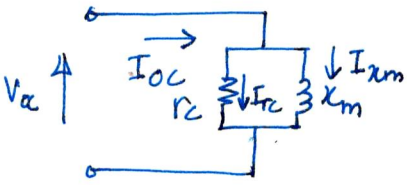
④ Transformer data: 50 MVA, 60Hz, 8000V/78000V

Open Circuit test (on LV side, ie 1° side)

$V_{oc} = 8000$V

$I_{oc} = 62.1$A

$P_{oc} = 206$ kW

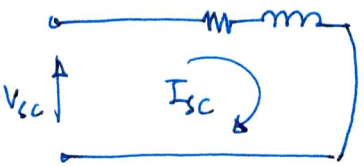


Short Circuit test (on LV side, 1° side)

$V_{sc} = 674$V

$I_{sc} = 6250$A

$P_{sc} = 187$ kW



From oc test data,

$$r_c = \frac{V_{oc}^2}{P_{oc}} = \frac{8000^2}{206 \times 10^3} = 310.6796\ \Omega$$

$$I_{rc} = \frac{V_{oc}}{r_c} = 25.75 \text{ A}$$

$$I_{xm} = \sqrt{I^2 - I_{rc}^2} = 56.5097 \text{ A}.$$

$\Rightarrow$ $x_m = \frac{V_{oc}}{I_{xm}} = 141.56\ \Omega$

Referred to LV side.

From SC Test data,

$$R = \frac{P_{sc}}{I_{sc}^2} = \frac{187 \times 10^3}{6250^2} = 4.7872\ m\Omega$$

$$Z = \frac{V_{sc}}{I_{sc}} = \frac{674}{6250} = 0.1078\ \Omega$$

$$X = \sqrt{Z^2 - R^2} = 0.1077\ \Omega$$

Referred to LV side

b) Refer every parameter to HV side

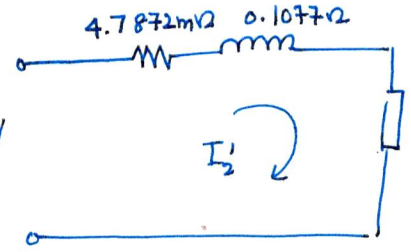
$$r_c' = r_c \cdot \left(\frac{78000^2}{8000^2}\right) = 310.6796 \times \frac{78000^2}{8000^2} = 29533.9795\ \Omega$$

$$x_m' = 141.56 \times \frac{78000^2}{8000^2} = 13457.0475\ \Omega$$

$$R' = 4.7872 \times 10^{-3} \times \frac{78000^2}{8000^2} = 0.4550\ \Omega$$

$$X' = 0.1077 \times \frac{78000^2}{8000^2} = 10.2382\ \Omega$$

(c) Equivalent circuit referred to LV side



Load current, I_2' = Rated current

$$= \frac{50 \times 10^6}{8000}$$

$$= 6250\ A.$$

$\cos\phi = 1.$

$$\Rightarrow \overline{I_2'} = 6250 \angle 0\ A.$$

$$|V_2'| = |V| - |I_2'|(R\cos\phi + X\sin\phi) = 8000 - 6250 \times 4.7872 \times 10^{-3} \times 1$$

$$= 7970.8\ V$$

$$\text{Voltage regulation}(\%) = \frac{8000 - 7970.8}{8000} \times 100 = 0.374\ \%$$

$$\text{Output power} = |V_2'|\,|I_2'|\cos\phi = 7970.8 \times 6250 = 49.8175\ MW$$

Iron loss (from OC Test) = P_{oc} = 206 kW.

Copper loss (at rated current) = 187 kW.

$\Rightarrow$ Input power = 49.8175 MW + 206 kW + 187 kW = 50.205 MW.

$\Rightarrow$ Efficiency = 99.217%.

d) At max. efficiency, $x^2 I_{fl}^2 R$ = Iron loss

$\Rightarrow$ Load factor (x) for max. $\eta = \sqrt{\frac{\text{Iron loss}}{\text{Copper loss}_{fl}}} = 1.05$

5) 20 kVA, 2400V / 240V transformer.

O.C Test : $V_{OC} = 240V$ $I_{OC} = 1\ A$ $P_{OC} = 126.6\ W$ (LV side)

S.C Test : $V_{SC} = 57.5V$ $I_{SC} = 8.34A$ $P_{SC} = 284\ W$. (HV side)

From OC Test, Iron loss = 126.6 W.

Rated current on HV side, $I_2 = \frac{20 \times 10^3}{2400} = 8.33\ A$

$\Rightarrow$ From SC Test, Copper loss at full load = 284 W

a) Given that load is such that current is full load current and voltage is rated terminal voltage at 0.8 pf lagging.

$\Rightarrow$ Output power $= |V_2||I_2|\cos\varphi = V_{rated} I_{rated} * 0.8$

$= 20 \times 10^3 \times 0.8 = 16000\ W$

Copper loss @ given load condition = Full load Cu. loss = 284 W.

Iron loss = 126.6 W

$\Rightarrow$ Input power = 16410.6 W

$\Rightarrow$ Efficiency, $\eta = 97.49\%$

b) Need to determine series branch parameters of equivalent circuit. (referred to HV side)

$R = \frac{P_{SC}}{I_{SC}^2} = \frac{284}{8.34^2} = 4.0833\ \Omega$

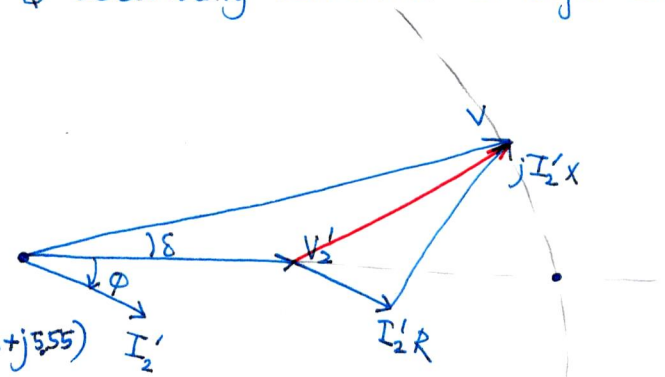
$Z = \frac{V_{SC}}{I_{SC}} = 6.8945\ \Omega$

$X = \sqrt{Z^2 - R^2} = 5.5554\ \Omega$

Load pf varied while load current & secondary terminal voltage constant

Regulation $\Rightarrow$ $|V|$ & $|V_2'|$

~~$|\bar{V}| - |\bar{V}_2'|$ is maximum when $\bar{V}$ is horizontal.~~ $|\bar{V} - \bar{V}_2'|$ is maximum when $\bar{V}$ is horizontal



$\Rightarrow$ $V\angle 0 = 2400\angle 0 + 8.34\angle\varphi\ (4.083 + j5.55)\ I_2'$

~~$2400\angle 0$~~

$= 2400\angle 0 + 8.34\angle\varphi * 6.89\angle 53.659°$

$\Rightarrow \varphi = -53.659°$

$\Rightarrow$ power factor @ max: regulation = $\cos(53.65) = 0.59$ lagging.

$|\bar{V}| = 2457.46\ V$

$\Rightarrow$ % Regulation = ~~2457.46~~ 2.3942 %